

8. (a) The table shows information about some Group 7 elements.

Element	Radius of the atom (nm)	Number of electrons in outer shell
chlorine	0.099	7
bromine	0.114	7
iodine	0.133	7

- (i) State how the radius of the atom changes down Group 7. [1]

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- (ii) Explain in terms of electronic structure why the reactivity of the elements decreases down Group 7. [2]

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- (b) Sea water contains sodium bromide, which is the raw material for the production of bromine.

Between 1953 and 2004 bromine, Br₂, was manufactured from sea water at a chemical plant in Amlwch, Anglesey.

Sea water was treated with chlorine, which converts sodium bromide into bromine.

Complete the balanced equation for the reaction between chlorine and sodium bromide. [3]



(c) 4.47 g of copper bromide was found to contain 1.27 g of copper.

(i) Calculate the mass of bromine in 4.47 g of the copper bromide. [1]

Mass = g

(ii) Calculate the simplest formula of the copper bromide.

You **must** show your working. [2]

$A_r(\text{Br}) = 80$ $A_r(\text{Cu}) = 63.5$

Simplest formula

